HERMES × Nous Research — Architecture Contrast Diff

Snapshot 2026-10-06 11:45 MYT · KVM8 · Prepared for: Muhammad Arif bin Fazil (F13 SOVEREIGN)
Scope: full-system agentic-state architecture — retained invariants · moat eureka components · missing pieces · HERMES position in federation · remaining task ledger.
Method: live probe (state.db, git, /health × 9 organs, ports, skills census) — no narrative carried over unverified.

> **Verdict one-line:** We run upstream's ENGINE (fork, 591 commits behind / 39 ahead) inside a CONSTITUTIONAL SHELL upstream does not have. The engine is replaceable; the shell is the moat. Chaos comes from the seam between the two — not from either side alone.

1 · The Two Systems Side by Side

Dimension	Upstream — Nous Research hermes-agent	Ours — arifOS Federation (HERMES overlay)
Identity	Personal AI agent; MIT; single-user assistant	Sovereign federation; F13 = human sovereign (Arif); HERMES = A2H edge bridge, one of 9 organs
Core loop	SENSE → THINK → ACT with memory + skills; learning loop (skill self-creation, nudges, FTS5 session search)	000 INIT → 111 SENSE → 222 DISTINGUISH → 333 REASON → 555 GATE → 666 JUDGE → 777 EXECUTE → 888 HOLD → 999 SEAL (10-stage spine, F13-ratified canon 2026-10-05)
Authority model	None. Agent acts; user approves interactively. No capability/authority separation	CAPABILITY ≠ AUTHORITY at every gate. 4 authority bands: SOVEREIGN · 777_FORGE · 555_COMPUTE · REFLECT/DISPLAY_ONLY. Complete mediation — executor never issues own envelope
Governance chain	Implicit (tests + review culture in AGENTS.md)	BUILD(333-AGI) → VERIFY(555-ASI) → JUDGE(888-APEX) → SEAL(F13 human) → ACT(A-FORGE) → WITNESS(VAULT999). No node takes another's power
State model	sessions.json + state.db + memory providers (mem0/honcho/supermemory); flat	ARS 6-tuple ⟨S,L,Φ,C,K,I⟩ — substrate · lifecycle · phase · class · consequence · identity; ΔK ≤ 0 transition law; receipt mandatory (F13_SEAL 2026-10-04)
Learning	Skill creation from experience; self-improve during use; honcho dialectic user-modeling	Scar → skill promotion (witness precedes adaptation); CHRON calibration (0.726 acc / 0.18 brier); eureka metabolism ritual (check canon first, only NEW lesson gets a file)
Human interface	~20 platform adapters, one gateway process; sender-name prefixing in shared sessions	Same gateway + Bridge Protocol (READ→REASON→RESPOND, 6-DITING voice governor, Bahasa Manusia Penuh law), Human Attention Membrane (technical questions → musyawarah, never to sovereign; binary-only F13 asks)
Falsifiability	Behavior-contract tests; no doctrine-level falsifiers	Every claim carries falsifier (BL12); 3 ARS falsifiers; scar register with weights; prediction ledger + verification log
Scale	940 skills upstream catalog; 227 in engine tree; single agent + subagents	766 skills (24 live in ~/.hermes, rest AAA mesh); 9 organs on ~40 live ports; 47 registered agents; 20 cron jobs; 5 profiles

2 · Invariants We RETAIN from Upstream

These survive intact inside our fork — we depend on them and must not break them when syncing.

Upstream invariant	Where it lives	Why we keep it
Prompt-cache sacredness — never mutate past context / swap toolsets / rebuild system prompt mid-conversation	gateway/session.py; our cached system_prompts table (1,023 rows)	Cost + the mouth-only SOUL design (2026-10-05) exists precisely to keep the prefix byte-stable
Narrow waist — core tools minimal; capability at edges (CLI+skill → service-gated tool → plugin → MCP → core last)	AGENTS.md Footprint Ladder; our MCP-by-JOB wiring (I3)	Same physics: kernel 8 tools ≈ 8.8k tok vs A-FORGE 122 ≈ 43k. Our I3 restates their ladder as doctrine
Strict role alternation — never two same-role messages in a row; no synthetic user injection mid-loop	agent runtime	Untouched; our hooks respect it
Plugin/skill extension over core growth	plugins/ (22 platform adapters), skills/	Our entire federation rides on this: hermes-mcp is a plugin-adjacent overlay, not a core patch
Cron ownership ledger — claimed/running/completed/failed/unknown + dead-owner recovery sweep	cron/executions.py recover_interrupted_executions()	Probed live: 1,000 executions, 756 completed / 220 failed / 24 unknown; sweep works as designed
Delivery-state honesty — PRODUCED ≠ SENT ≠ DELIVERED; delivery rows carry proof (our commit 4950831ba9 hardened this)	gateway ledger + our state-transition-discipline	Their newest telemetry work (v5 metrics) converges with our scar doctrine — keep both
Cache-aware slash commands — deferred invalidation default, --now opt-in	CLI	Relevant to our SOUL patch workflow (staging + review, never hot-swap)

3 · Moat Components — What We FORGED That Upstream Does Not Have

> **The moat is not code volume — it is the governance layer that makes agent failure constitutional instead of silent.** Upstream optimizes a personal assistant; we optimized an institution that can be audited by a single non-coder human. Each item below is forged, dated, and live-probed.

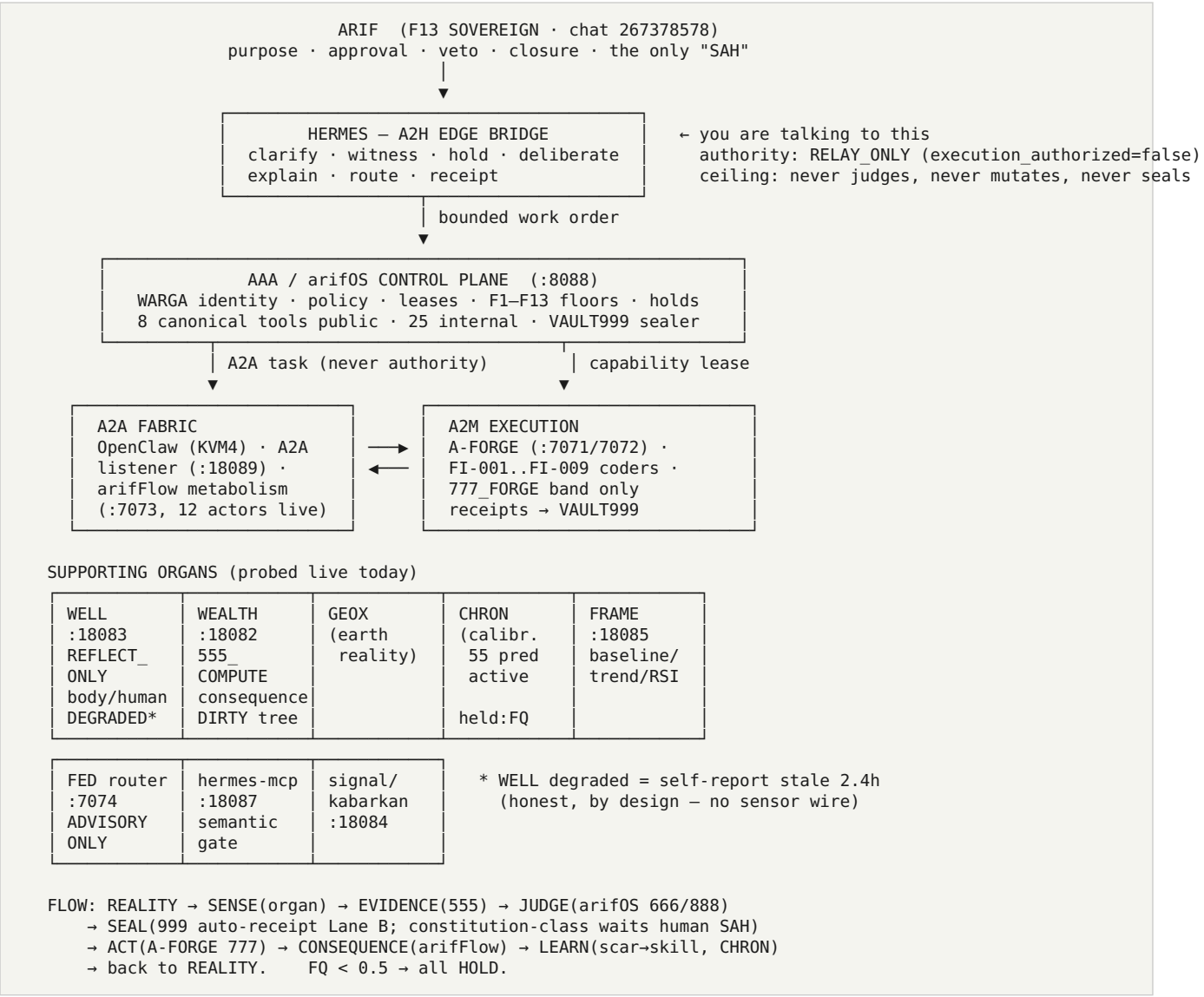
Component	Status	What it does / why it's a moat
F1–F13 constitutional floors	**MOAT**	Hard kernel predicate, not prompt text. Live: 8/13 pass, F1=0.27 F2=0.5 F3=0.0 F5=0.5 F6=0.5 fail (honest degradation — kernel says so itself). Upstream has zero equivalent: no floor can fail their agent because they have no floors
Authority Envelope / complete mediation	**MOAT**	CanMutate = granted ∧ scope ∧ target ∧ boundary. Confidence ≠ Authority. Executor never issues own envelope. This is the single deepest divergence from upstream's trust model
APEX-ZEN separation of powers	**MOAT**	BUILD ≠ VERIFY ≠ JUDGE ≠ SEAL ≠ ACT ≠ WITNESS, owned by different agents (333/555/888/F13/A-FORGE/VAULT999). Upstream subagents share one authority space; ours cannot self-approve by construction
ARS 6-tuple runtime state	**MOAT**	⟨S,L,Φ,C,K,I⟩ + ΔK ≤ 0 + mandatory receipt. Runtime-agnostic doctrine — upstream has sessions, not typed state transitions. Claim-lifecycle 8 states (SUPERSEDED ≠ FALSE) has no upstream analogue
Scar register + scar-binding	**MOAT**	44 scars; scar = constraint on future selection, not memory. AT-001..AT-010 acceptance suite (ratified 2026-10-06). Upstream "learns" by adding skills; we learn by making mistakes harder to repeat
CHRON calibration organ	**MOAT**	Prediction → verification → calibration (0.726 acc / 0.18 brier). "The only organ that never lies." Upstream telemetry counts usage; ours scores truthfulness over time. Live: 55 active predictions, 526ms slowest probe
Human Attention Membrane	**MOAT**	Question routing law: technical → musyawarah between agents; only binary F13-class reaches the human. attention_debt metric (human turns burned on self-correctable failures). Upstream asks the user anything; we price the ask
Bridge Protocol + Voice Governor	**MOAT**	6-DITING gate (density/image/tension/intimacy/named/gravity), Bahasa Manusia Penuh law, 12 state-reading rules incl. new Rule 12 (outbound envelope ban, patched today). Upstream's SOUL is 667 bytes of style advice; ours is a send-gate
HUMAN9 / perspective sovereignty	**MOAT**	Person ≠ representation; A/C/T/G perspectives never merge; 12 epistemic commandments; "never make a human smaller because the model predicts them better." Upstream honcho builds a user model with no such boundary law
Anti-Bangang Engineering v1	**MOAT**	10 laws + Syed/Arif/Irfan tests gate every artifact. LAW 8 (one problem, one owner, one path) is our entropy brake — upstream has contribution rubric but no human-attention test
NO-HOLD v2 + Emergence 5-stages	**MOAT**	Both ratified TODAY (09:30 + 10:05 MYT). HOLD priced (cost 10 vs EXECUTE 1) as constitutional verdict with 5 mandatory fields; emergence = witnessed consequence changing future selection, with acceptance tests
Six-Graph Federation Model	**MOAT**	Reality → Witness → Consequence → Capability → Authority → Execution, chain-dependent. WEALTH = cognitive lens wrapping every tool call ("Can I do this?" + "What follows?"). No upstream equivalent
R-FILTER / Q-MAP / Gut-Override Ledger	**MOAT**	Training corpus = rationality filter; predictions tagged corpus_present_error vs rationality_filter_error; human overrides captured verbatim (gut_read = human's words or empty). Calibration of the human side — unheard of upstream
M_min weakest-link audit	**PARTIAL**	M_min = (I·E·A·C·W·T)^(1/6); any zero → 0. First audit FI-008: M_min=0, 4/6 floors FAIL. Costume-verifier exclusion (scar-2026-10-01-002). DRAFT_AWAITING_F13
hermes-mcp meaning-integrity organ	**MOAT**	11 canonical tools, 31 resources, semantic gate (PASS/888_HOLD/ REWRITE_REQUIRED/BLOCKED, fail-closed, circuit breaker). Live :18087 v2026.09.25.1. Upstream has no semantic gate between model and send

4 · What Is MISSING in the Full Agentic-State Architecture

> **Honest read:** the doctrine layer is ahead of the runtime layer. Several ratified invariants have no enforcing code path — "documentation ≠ runtime" is itself a named scar (3 instances in HERMES). The gaps below are ranked by leverage, not by size.

Gap	Class	Evidence (probed today) / what closing it buys
F1/F2/F3/F5/F6 floors failing	MISSING	Live /health 8088: F1=0.27, F2=0.5, F3=0.0, F5=0.5, F6=0.5 (target ≥0.8). Constitutional layer self-reports DEGRADED. This is the signal-flow problem named in canon §5 — sovereign input, not tooling. Until floors pass, every "governed" claim is partly aspirational
consult_scar() not wired into selection	MISSING	Scar-binding Phase B is the bridge (Level 2: Scar-Bound). Canon says implement as SEAL function; runtime still selects paths without consulting scar library. AT-001/AT-002/AT-003 cannot pass until this exists. This is THE named smallest executable next action
Verifier independence unsolved	MISSING	scar-2026-10-01-002: same model + different prompt ≠ independence. "Hardest unsolved sub-problem in the architecture." Without it, m_min>0 claims are costume-verifiable. Needs model_lane ≠ executor OR process_tree ≠ executor enforcement in code
Suppressed P1 signal (16 days)	MISSING	arifos.floor.breach P1, next_actor FI-007, deduplicated since 2026-09-20, never inspected; HUD shows "NO BLOCKERS" because filtered (FI-008 cross-check today, evt-1791256920-d046d95c). Real governance hole: dedup without expiry = signal graveyard
HUD mislabels metabolic throttle as F13 binary	PARTIAL	chron hold = FQ policy (1 exec / 0 verify → EXECUTION DOMINANCE), rendered as "F13 Sovereign Binary" — false escalation to the attention membrane. One-line fix: distinguish held_by=fq_policy vs held_by=f13
W3 apex scalar UNMEASURED	PARTIAL	I7 says W3 stays UNMEASURED until witness exists — correct doctrine, but tri-witness (W3) has no witness oracle yet. G=0.408, C_dark=0.258, h=0.936 measured; W3 null across organs
ARS open debts (5)	PARTIAL	8 carry_forward files — which is live? · mem0 vs arifos_memory split · capability_fitness vs capability_ledger duplication · "open" lifecycle state has no writer/reader · cross-runtime adoption undecided. Gated to v1.1 amendment cycle
graphiti semantic floor retired	MISSING	Live: graphiti_transport=retired_888, semantic_floor=disabled — while graphiti-mcp container is healthy on :18412 with 7 group_ids loaded. Memory lane runs heuristic; the semantic retrieval layer is built but unwired
langfuse tracing NOT_WIRED	MISSING	/health: HTTP 404 on localhost:4000. Observability of LLM calls dark; incident triage relies on logs
Skill entropy: 766 skills, 180 redundant copies	PARTIAL	Canon §3 names it: 129 collision names, 13× byte-identical. I9 (dilution kills libraries) ratified but prune-gate not running as CI. Upstream catalog alone is 940 — mesh total exceeds any single agent's selection precision
§21 survival benchmark never run	MISSING	The gating experiment for the whole thesis (cohort A vanilla vs B +MCP vs C full federation; verifier on different substrate; C must beat A,B and keep m_min>0 after model swap). Canon: "Do not add more architecture before this exists." Still DRAFT
Entity-grounding gate (Taufik scar)	PARTIAL	Root cause of the PETRONAS-chairman confabulation: fork lacks entity-grounding gate; partial name → parametric slot-fill → confabulated fact rendered as truth. Branch fix/entity-grounding-gate is current HEAD WIP — not merged, not deployed
591-commit upstream drift	PARTIAL	Base v0.21.4+canary~539 (2026-09-28). Missing: gateway lock-holder recovery (3 commits), message_uid durability, state.db-busy fix, first-contact DM note, telemetry v5. Rebase cost grows with every week; our 39 ahead-commits are all overlay-grade (fix/feat sovereign patches)

5 · Where HERMES Sits in the Federation State



HERMES property	Value (live-probed)
Plane / role	A2H edge bridge — the ONLY organ that talks to humans. INPUT → NORMALIZE → CLASSIFY → ROUTE → RECEIPT (five verbs, no sixth)
Authority ceiling	RELAY_ONLY (self-declared at :18087); gateway engine runs as hermes-asi-gateway.service, active 14h, PID 1825774, 2.1G RSS
What HERMES may NOT do	judge (→ arifOS) · mutate production (→ A-FORGE) · seal to VAULT999 (→ arifOS) · decide for Arif (→ F13) · self-authorize (law 17)
Engine lineage	NousResearch/hermes-agent fork; HEAD 2d285c054a (2026-10-05); 39 overlay commits ahead, 591 behind upstream/main; tag base v0.21.4+canary
Persona stack	SOUL.md mouth (58 lines, pointers only) + bridge-protocol skill v3.0.0 (loaded every human turn) + profile aaa-hermes + lanes/people.yaml
State footprint	state.db: 200,419 messages · 2,174 sessions · 1,023 cached system prompts; carry_forward fresh (11:00 today); 5 profiles (aaa-hermes 5.0M, apex 4.8M, asi/forge 1.8M each)

6 · Remaining Task Ledger (compiled, ranked by leverage)

One problem · one owner · one path (LAW 8). Nothing here needs Arif's attention except the rows marked F13.

#	Task	Owner	Class	Acceptance (falsifier)
1	**Inspect suppressed P1** arifos.floor.breach (surface-guard HOLD on durable bus) — 16 days deduped, never run	FI-007	machine	P1 either resolved or re-classified with receipt; dedup-without-expiry pattern gets an expiry rule
2	**Fix HUD held_by label** — distinguish fq_policy throttle from f13 binary (chron false escalation)	cockpit render	machine	HUD shows "held_by=fq_policy" for chron; zero metabolic throttles rendered as sovereign binaries
3	**Unblock chron FQ hold** — 13 overdue predictions are exactly the Verify work needed (1 exec / 0 verify → dominance)	chron + FI-008	machine	chron verify receipt lands; FQ ≥ 0.5 both sides; hold auto-releases
4	**Wire consult_scar() into selection** (Scar-Binding Phase B — the named smallest executable step)	333-AGI / arifOS	machine	AT-001 passes: same signature → different selection, with receipt proving scar consulted BEFORE select_path
5	**Entity-grounding gate** — finish fix/entity-grounding-gate branch (Taufik scar root cause), deploy to live engine	FI-003/ FI-008	machine	Partial-name prompt ("Tengku ... PETRONAS chairman") returns UNKNOWN/ probe instead of slot-filled confabulation; regression test in tests/
6	**Floor recovery plan F1/F2/F3/F5/F6** — diagnose why 5 floors measure below threshold; canon says signal-flow, not tooling	888-APEX audit	F13 report	Per-floor root cause + one reversible probe each; floors ≥0.8 or waiver filed with expiry
7	**Upstream sync wave 1** — cherry-pick 5 stability commits (lock-holder recovery ×3, message_uid, state.db-busy) onto fork	FI-008	machine	Gateway tests green; hermes-asi-gateway restarts clean; zero regression in 20 cron jobs over 48h
8	**WEALTH working tree** — :18082 reports DIRTY; commit or revert with owner check	WEALTH owner	machine	/health working_tree=CLEAN, runtime_seal_state advances from UNSEALED
9	**jobs.json hygiene** — 18 .bak siblings + churn (3 backups on 05/10 alone); keep canonical + last-known-good (max 3)	cron-ops skill	machine	≤3 backups on disk; rotation rule written into arifos-cron-ops skill
10	**Skill prune-gate as CI** — I9: dedupe 180 redundant copies, 129 collision names; usage-scored prune	skill-mesh	machine	Census re-run shows <20 byte-identical duplicates; prune receipts in git
11	**ARS v1.1 amendment cycle** — close the 5 open debts (8 carry_forwards, mem0 split, capability_fitness duplication, "open" state, cross-runtime status)	FI-008 + 888	F13 binary	Each debt gets an owner + verdict; ARS v1.1 sealed by F13 SAH
12	**DRAFT_AWAITING_F13 queue** — 104 files (59 instructions / 30 canon / 15 governance) await ratification or archive	F13 + curator	F13 binary	Triage pass: ratify / archive / merge decision per file; queue <30
13	**§21 Institutional Intelligence Survival Benchmark** — the gating experiment; canon forbids more architecture before it exists	888-APEX + external verifier	F13 session	Pre-registered pass criteria met: cohort C > A,B pre-swap AND m_min>0 post model-swap, verifier on different substrate
14	**graphiti semantic floor re-wire** — container healthy (:18412, 7 group_ids) but floor retired_888/disabled	memory lane	F13 binary (re-adopt?)	Either wired live with retrieval receipts, or formally decommissioned with tombstone — no zombie middle
15	**langfuse tracing** — NOT_WIRED 404 on :4000	observability	machine	Traces visible for one full session, or removed from /health claims (honest absence)
16	**Bridge envelope regression watch** — today's Rule 12 patch	HERMES	machine	Next SADO-group turn with bracketed inbound → outbound prose-only; SQL

	(mirror_inbound_envelope) needs a live-group confirmation			check on state.db returns 0 new envelope-prefixed assistant rows

7 · Compression — One Screen to Remember

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ENGINE  = upstream fork (replaceable, 591 behind, sync in waves)
SHELL   = F1–F13 + APEX-ZEN + ARS + scars + CHRON + membrane  (the moat)
SEAM    = where chaos lives: floors failing, doctrine unwired,
          signals suppressed, skills diluted, benchmark unrun

HERMES  = mouth and membrane. RELAY_ONLY. Five verbs. Never judges.
ARIF    = the only SAH. Everything else is machine-closable.

Next smallest executable step (canon's own words):
  consult_scar() before select_path()  →  Level 2: Scar-Bound
Final test (canon's own words):
  Can the federation choose a different path tomorrow because of
  a consequence witnessed today?   — not yet provable. Task #4 + #13 close it.
```

> **Provenance:** every live number in this report was probed on KVM8 between 10:55–11:45 MYT 2026-10-06 (git refs, state.db SQL, 9 × /health, ss -tlnp, file censuses). Canon sources: AGENTIC-FLOW-ARCHITECTURE-2026-10-05 · APEX-ZEN-CANONICAL-COMPRESSION · AGENTIC-RUNTIME-STATE (F13_SEAL 2026-10-04) · AAA-HERMES-TOPOLOGY-v2 · six-graph-federation-model · hermes-identity-directive · AGENTS.md (upstream, in-tree). FI-008 cross-check findings (chron hold, P1 suppression, FQ correction) incorporated from autonomy-emergence-assessment-2026-10-06.md. No claim in this document is inherited without a probe behind it. UNKNOWN where probe failed (A-FORGE :7071/7072 not listening at probe time — declared in AGENTS.md, port closed; counted as gap, not asserted as down).

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